Comprehensive Report: Cross-Layer Simulation for Latency-Sensitive Traffic Prioritization in Wi-Fi Networks

Abstract

This report details a comprehensive cross-layer simulation study designed to analyze the complex interactions between different network layers in IEEE 802.11ac wireless environments. The simulation utilizes NS-3.44 to model various traffic classes, TCP variants, propagation models, and mobility scenarios under configured QoS parameters.

Contents

Chapter 1

Introduction

The increasing demand for low-latency applications in wireless networks necessitates a deeper understanding of cross-layer interactions. This simulation study investigates latency-sensitive traffic prioritization in Wi-Fi networks using a comprehensive cross-layer approach, examining how different network layers interact and affect overall performance.

Chapter 2

Simulation Setup and Parameters

2.1 Platform Configuration

- **Platform**: NS-3.44
- **WiFi Standard**: IEEE 802.11ac
- **Frequency**: 5.0 GHz (5.0e9)
- **Channel Bandwidth**: 20 MHz (default, ns-3 defaults)
- **TX Power**: 20 dBm (configurable)
- **MCS Levels**: 0-9

2.2 QoS Parameters

2.2.1 802.11e QoS/WMM (EDCA/WMM)

Parameter	**Value**	**Description**
Queue Management	FqCoDelQueueDisc	Fair Queueing with CoDel AQM
AP Queue Size	200 packets	Maximum queue size at AP
WiFi MAC Queue Size	600 packets	Per-AC queue size
Target Queue Delay	5ms	CoDel target delay
Interval	50ms	CoDel interval

Table 2.1: QoS Configuration Parameters

2.2.2 Traffic Access Category Mapping

2.3 TCP Variants

- TcpBbr (Default)

Traffic Class	Priority	AC Category	Data Rate (Mbps)	Interval (ms)	UDP/TC
TC_VO (voice)	46	AC_VO	10	0.5	UDP
TC_VI (video)	34	AC_VI	50	2	UDP
TC_BE (Best-Effort)	0	AC_BE	200	5	TCP
TC_BK (Background)	8	AC_BK	1000	5	TCP

Table 2.2: Traffic Class to Access Category Mapping

- TcpCubic (Default)
- TcpNewReno (Default)

2.4 Propagation Models

2.4.1 LogDistance Model

- **Reference Distance**: 1.0 meter
- **Reference Loss**: 46.6777 dB
- **Path Loss Exponent**: Configurable (default 3.0)

2.5 Mobility Models

2.5.1 RandomWalk2dMobilityModel

- **Bounds**: Rectangle (-mobilityRange, mobilityRange, -mobilityRange, mobilityRange)
- **Distance**: 8.0 meters
- **Speed**: ConstantRandomVariable (configurable, default 2.0 m/s)
- **Application**: Station 1 and Station 2

2.5.2 ConstantPositionMobilityModel

- **Application**: AP and traffic servers (static positioning)

2.6 TCP Configurations

2.7 Simulation Timings

2.8 Error Model

- **Model**: NIST Error Rate Model

Parameter	Value	Description
TCP Variant	TcpBbr (configurable)	Congestion control algorithm
Segment Size	1448 bytes	TCP maximum segment size
Initial CWND	10 segments	Initial congestion window
Send/Receive Buffer	256 KB	Socket buffer sizes
Data Rate	10 Mbps	Target sending rate for TCP flows

Table 2.3: TCP Configuration Parameters

Parameter	Value	Description
Simulation Time	300s (configurable)	Main simulation duration
Warmup Time	10s (configurable)	Initial stabilization period
Monitoring Interval	100ms	Queue sampling interval
Cross-layer Sampling	500ms	Cross-layer correlation sampling
Throughput Sampling	1s	Throughput measurement interval

Table 2.4: Simulation Timing Parameters

- **Purpose**: PHY layer error modeling
- **Minimum error rate**: 0.001 (0.1%)

2.9 Queueing Theory Parameters

- **CoDel Parameters**: Target=5ms, Interval=50ms
- **Queue Sampling**: Records queue lengths over time for each AC
- **Loss Rate Calculation**: Loss Rate = Drops / (Successes + Drops)

Chapter 3

Assumptions and Analytical Parameters

3.1 Network Assumptions

- **Single AP**: One access point serves two stations
- **Wired Backhaul**: AP connected to servers via 1Gbps links with 2ms delay
- **Interference-Free**: No external interference modeled
- **Perfect Synchronization**: No clock drift between nodes

3.2 Traffic Patterns

- **Constant Bit Rate (CBR) UDP**: VO at 3.2ms intervals, VI at 4.8ms intervals
- **Bulk TCP Transfers**: BE and BK use TCP with 10Mbps target rate
- **Independent Flows**: No interaction between traffic generators
- **Infinite Source Data**: TCP sources never run out of data to send

3.3 QoS Implementation

- **802.11e EDCA**: Implemented via ns3::QosTxop
- **DSCP to UP Mapping**: Uses standard DSCP-to-802.11 UP mapping
- **Per-AC Queues**: Separate queues for each access category
- **AP-side Queue Management**: FQ-CoDel at AP, no AQM at stations

3.4 Measurement Assumptions

- **Steady-State Analysis**: Warmup period of 10s to reach steady state
- **Sufficient Samples**: Assumes enough packets for statistical significance

3.5 Cross-Layer Correlation Analysis Framework

The following metrics will be analyzed for correlation:

1. **Queue Length vs RTT/Latency**: Impact of congestion on delay
2. **Queue Length vs CWND**: Impact of congestion on TCP window
3. **PHY Loss Rate vs RTT/Latency**: Impact of wireless errors on delay
4. **PHY Loss Rate vs CWND**: Impact of wireless errors on TCP behavior

3.5.1 Correlation Interpretation Guidelines

Correlation Value	**Interpretation**
> 0.7	Strong positive correlation
$0.3 - 0.7$	Moderate positive correlation
$-0.3 - 0.3$	Weak/No correlation
$-0.7 - -0.3$	Moderate negative correlation
< -0.7	Strong negative correlation

Table 3.1: Correlation Coefficient Interpretation

Chapter 4

Test and Results

4.1 Simulation Configuration Parameters

The following parameters were configured for the test simulations:

```
// Simulation timing and configuration
double simulationTime = 300.0;
double warmupTime = 10.0;
bool enableQos = true;

// PHY layer parameters
double txPower = 20.0;
double noiseFigure = 7.0;
std::string errorRateModel = "Nist";

// TCP parameters
std::string tcpVariant = "TcpNewReno";
uint32_t tcpSegmentSize = 1448;

// Queue parameters
uint32_t wifiMacQueueSize = 600;
uint32_t apQueueSize = 200;

// Station distances
double sta1Distance = 30.0;
double sta2Distance = 30.0;

// Propagation parameters
double pathLossExponent = 3.0;

// Mobility parameters
bool enableMobility = false;
double mobilitySpeed = 1.5;
double mobilityDistance = 10.0;
double mobilityBoundsX = 100.0;
```

```
double mobilityBoundsY = 100.0;

// Monitoring parameters
double monitorIntervalMs = 100.0;
double crossLayerSampleIntervalMs = 500.0;
double throughputSampleIntervalSec = 1.0;

// Output configuration
std::string outputPrefix = "qos_analysis";
double progressInterval = 10.0;
```

4.2 Mobility Analysis

This section analyzes the impact of station mobility on QoS performance across different mobility levels.

4.2.1 TCP BBR Throughput and RTT Analysis

Metric	Static (0 m/s)	Low Mobility (1 m/s)	Medium Mobility (3 m/s)
BE Throughput	11.52 Mbps	11.52 Mbps	11.52 Mbps
BK Throughput	11.52 Mbps	11.52 Mbps	11.52 Mbps
BE Avg RTT	5.08 ms	5.17 ms	5.17 ms
BK Avg RTT	5.32 ms	6.88 ms	6.10 ms
BE Max RTT	9.82 ms	14.21 ms	19.61 ms
BK Max RTT	19.28 ms	69.31 ms	54.44 ms
BE Retrans	31	32	43
BK Retrans	0	5	4

Table 4.1: TCP BBR Performance Under Different Mobility Conditions

4.2.2 TCP Congestion Window Analysis

Metric	Static (0 m/s)	Low Mobility (1 m/s)	Medium Mobility (3 m/s)
BE Avg CWND	20,360 bytes	22,051 bytes	21,584 bytes
BK Avg CWND	25,986 bytes	25,821 bytes	25,459 bytes
BE Max CWND	102,354 bytes	79,057 bytes	92,489 bytes
BK Max CWND	37,946 bytes	53,511 bytes	46,180 bytes

Table 4.2: TCP Congestion Window Behavior Under Mobility

4.2.3 Key Observations

- **Throughput Stability**: Both TCP flows maintain consistent 11.52 Mbps throughput across all mobility levels, demonstrating robust performance under varying channel conditions.
- **RTT Impact**: Background (BK) traffic shows greater sensitivity to mobility with a 14.7% RTT increase compared to only 1.77% for Best-Effort (BE) traffic.
- **Retransmission Increase**: BE retransmissions increase by 39% with mobility, indicating higher packet loss under mobile conditions.
- **CWND Behavior**: BE traffic shows increased average congestion window size under mobility, while BK shows decreased average CWND, suggesting different congestion control responses.
- **Maximum RTT Spikes**: Significant increases in maximum RTT values are observed with mobility, particularly for BK traffic (69.31 ms at low mobility vs 19.28 ms static).

4.3 TCP Variant Analysis Comparison

4.3.1 Experiment Setup Summary

Parameter	**Value**
Simulation Duration	300 seconds
WiFi Standard	802.11ac
Frequency	5 GHz
Channel Bandwidth	80 MHz
Rate Control	MinstrelHT (MCS 0-9)
Error Model	Nist
Station Distance	25 meters (both stations)
TCP Variants Tested	NewReno, CUBIC, BBR

Table 4.3: TCP Variant Analysis Experiment Setup

4.3.2 TCP Throughput and Congestion Metrics

4.3.3 UDP Latency Impact

4.3.4 PHY Layer Statistics

4.3.5 Key Findings & Analysis

Metric	TCP NewReno	TCP CUBIC	TCP BBR
Throughput (BE)	11.52 Mbps	11.52 Mbps	11.52 Mbps
Throughput (BK)	11.52 Mbps	11.52 Mbps	11.52 Mbps
Avg RTT (BE)	5.07 ms	5.07 ms	5.08 ms
Avg RTT (BK)	5.28 ms	5.28 ms	5.32 ms
Avg CWND (BE)	36.2 MB	27.5 KB	20.4 KB
Avg CWND (BK)	36.2 MB	24.6 KB	26.0 KB
Max CWND (BE)	72.4 MB	40.5 KB	102.4 KB
Max CWND (BK)	72.4 MB	34.8 KB	37.9 KB
Retransmissions (BE)	0	0	31
Retransmissions (BK)	0	0	0

Table 4.4: TCP Variant Performance Comparison

Metric	TCP NewReno	TCP CUBIC	TCP BBR
VO Mean Latency	2.101 ms	2.101 ms	2.101 ms
VI Mean Latency	2.210 ms	2.209 ms	2.211 ms
VO P99 Latency	2.569 ms	2.562 ms	2.562 ms
VI P99 Latency	2.707 ms	2.702 ms	2.704 ms
VO Jitter	0.011 ms	0.012 ms	0.025 ms
VI Jitter	0.116 ms	0.130 ms	0.136 ms
UDP Packet Loss	0.00%	0.00%	0.00%

Table 4.5: UDP Traffic Impact Across TCP Variants

Metric	TCP NewReno	TCP CUBIC	TCP BBR
TX Attempts	1,218,752	1,218,530	1,218,416
RX Success	1,060,692	1,060,671	1,059,849
PHY Loss Rate	1.44%	1.42%	1.43%

Table 4.6: PHY Layer Performance Across TCP Variants

1. **Congestion Window Behavior**
   - **NewReno**: Shows extremely large CWND (tens of MB) - likely unrealistic for WiFi or indicates bufferbloat
   - **CUBIC**: Moderate CWND sizes (tens of KB) with stable operation
   - **BBR**: Variable CWND with highest retransmissions for BE traffic (31 vs 0 for others)

2. **Impact on UDP Traffic**
   - **Latency**: Nearly identical across all variants (2.1ms VO, 2.2ms VI)
   - **Jitter**: BBR shows 2x higher jitter for VO traffic (0.025ms vs 0.011-0.012ms)
   - **Priority Preservation**: All variants maintain QoS differentiation (VO ¡ VI latency)

3. **Retransmission Patterns**
   - **NewReno & CUBIC**: Zero retransmissions (possibly due to large buffers)
   - **BBR**: 31 retransmissions for BE traffic, suggesting more aggressive probing
   - This aligns with BBR's design: faster probing can cause more losses in congested WiFi
4. **Cross-Layer Insights**
   - **NewReno**: Shows strong positive correlation between loss and CWND (0.704)
   - **CUBIC**: Moderate queue-RTT correlation for BK traffic (0.330)
   - **BBR**: Shows weak correlations, suggesting different congestion control dynamics
5. **Bufferbloat Indication (NewReno)**
   - The massive CWND in NewReno (72MB max) suggests severe bufferbloat
   - Despite this, UDP latency remains low, indicating effective QoS prioritization
   - This could be problematic in real deployments with limited buffer space

4.4 Cross-Layer Correlation Analysis Results

4.4.1 Correlation Strength by Mobility Level

Correlation Type	Static (0 m/s)	Low Mobility (1 m/s)	Medium Mobility (3 m/s)
BE Queue-RTT	0.009 (None)	0.141 (None)	0.055 (None)
BE Loss-CWND	-0.008 (None)	0.391 (Moderate)	0.041 (None)
BK Queue-RTT	-0.034 (None)	0.592 (Moderate)	0.353 (Moderate)
BK Loss-CWND	0.041 (None)	0.127 (None)	-0.067 (None)
VO Queue-Latency	0.000 (None)	0.000 (None)	0.000 (None)
VO Loss-Latency	0.033 (None)	0.107 (None)	-0.013 (None)
VI Queue-Latency	0.063 (None)	-0.009 (None)	0.073 (None)
VI Loss-Latency	-0.013 (None)	0.202 (Weak)	0.146 (None)

Table 4.7: Cross-Layer Correlation Analysis Under Different Mobility Conditions

4.4.2 Correlation Analysis by TCP Variant

4.4.3 Key Observations from Correlation Analysis

- **Weak Overall Correlations**: Most cross-layer correlations remain weak (¡0.3) across all mobility levels and TCP variants, indicating complex, non-linear relationships between layers.
- **BK Sensitivity**: Background traffic shows moderate queue-RTT correlation with

Correlation	TCP NewReno	TCP CUBIC	TCP BBR
BE Loss-CWND	0.704 (Strong)	0.219 (Weak)	-0.008 (Weak)
BK Loss-CWND	0.704 (Strong)	0.287 (Weak)	0.041 (Weak)
BE Queue-RTT	-0.032 (Weak)	0.002 (Weak)	0.009 (Weak)
BK Queue-RTT	-0.035 (Weak)	0.330 (Moderate)	-0.034 (Weak)

Table 4.8: Cross-Layer Correlation Analysis by TCP Variant

mobility (0.592 at 1 m/s), suggesting that queue dynamics more directly impact latency for lower-priority traffic under mobile conditions.

- **VO Stability**: Voice traffic correlations remain near zero regardless of mobility or TCP variant, demonstrating the effectiveness of QoS prioritization in isolating high-priority traffic from cross-layer interactions.
- **NewReno Anomaly**: TCP NewReno shows strong correlations (0.704) between loss and congestion window, likely due to its traditional loss-based congestion control mechanism.
- **Correlation Instability**: Some correlations show non-linear patterns with mobility, suggesting threshold effects in cross-layer interactions.
- **BBR Uniqueness**: BBR's weak correlations across all metrics suggest fundamentally different congestion control dynamics compared to loss-based algorithms.

4.5 Impact Analysis by Traffic Class

Traffic Class	Primary Impact	Sensitivity	QoS Compliance
VO (Highest)	Minimal latency increase, zero loss	Low	Excellent (P99 ¡ 2.61ms)
VI (High)	Moderate jitter increase, zero loss	Medium	Excellent (P99 ¡ 2.87ms)
BE (Medium)	Increased retransmissions, RTT spikes	High	Throughput maintained
BK (Low)	RTT variability, occasional retrans	High	Throughput maintained

Table 4.9: Traffic Class Performance Summary

Parameter	Value
Simulation Duration	300 seconds
WiFi Standard	802.11ac
Frequency	5 GHz
Channel Bandwidth	80 MHz
Rate Control	MinstrelHT (MCS 0-9)
Error Model	Nist
Station Distance	25 meters (both stations)
TCP Variants Tested	NewReno, CUBIC, BBR

Table 4.10: TCP Variant Analysis Experiment Setup

Metric	TCP NewReno	TCP CUBIC	TCP BBR
Throughput (BE)	11.52 Mbps	11.52 Mbps	11.52 Mbps
Throughput (BK)	11.52 Mbps	11.52 Mbps	11.52 Mbps
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Max CWND (BE)	72.4 MB	40.5 KB	102.4 KB
Max CWND (BK)	72.4 MB	34.8 KB	37.9 KB
Retransmissions (BE)	0	0	31
Retransmissions (BK)	0	0	0

Table 4.11: TCP Variant Performance Comparison

Metric	TCP NewReno	TCP CUBIC	TCP BBR
VO Mean Latency	2.101 ms	2.101 ms	2.101 ms
VI Mean Latency	2.210 ms	2.209 ms	2.211 ms
VO P99 Latency	2.569 ms	2.562 ms	2.562 ms
VI P99 Latency	2.707 ms	2.702 ms	2.704 ms
VO Jitter	0.011 ms	0.012 ms	0.025 ms
VI Jitter	0.116 ms	0.130 ms	0.136 ms
UDP Packet Loss	0.00%	0.00%	0.00%

Table 4.12: UDP Traffic Impact Across TCP Variants

Metric	TCP NewReno	TCP CUBIC	TCP BBR
TX Attempts	1,218,752	1,218,530	1,218,416
RX Success	1,060,692	1,060,671	1,059,849
PHY Loss Rate	1.44%	1.42%	1.43%

Table 4.13: PHY Layer Performance Across TCP Variants

4.6 TCP Variant Analysis Comparison

4.6.1 Experiment Setup Summary

4.6.2 TCP Throughput and Congestion Metrics

4.6.3 UDP Latency Impact

4.6.4 PHY Layer Statistics

4.6.5 Cross-Layer Correlations by TCP Variant

Correlation	TCP NewReno	TCP CUBIC	TCP BBR
BE Loss-CWND	0.704 (Strong)	0.219 (Weak)	-0.008 (Weak)
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BE Queue-RTT	-0.032 (Weak)	0.002 (Weak)	0.009 (Weak)
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Table 4.14: Cross-Layer Correlation Analysis by TCP Variant

4.6.6 Key Findings & Analysis

1. Congestion Window Behavior

- **NewReno**: Shows extremely large CWND (tens of MB) - likely unrealistic for WiFi or indicates bufferbloat
- **CUBIC**: Moderate CWND sizes (tens of KB) with stable operation
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2. Impact on UDP Traffic

- **Latency**: Nearly identical across all variants (2.1ms VO, 2.2ms VI)
- **Jitter**: BBR shows 2x higher jitter for VO traffic (0.025ms vs 0.011-0.012ms)
- **Priority Preservation**: All variants maintain QoS differentiation (VO ¡ VI latency)

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- **NewReno & CUBIC**: Zero retransmissions (possibly due to large buffers)
- **BBR**: 31 retransmissions for BE traffic, suggesting more aggressive probing
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4. **Cross-Layer Insights**
   - **NewReno**: Shows strong positive correlation between loss and CWND (0.704)
   - **CUBIC**: Moderate queue-RTT correlation for BK traffic (0.330)
   - **BBR**: Shows weak correlations, suggesting different congestion control dynamics
5. **Bufferbloat Indication (NewReno)**
   - The massive CWND in NewReno (72MB max) suggests severe bufferbloat
   - Despite this, UDP latency remains low, indicating effective QoS prioritization
   - This could be problematic in real deployments with limited buffer space
6. **Recommendations for Mixed Traffic Environments**

Analysis Summary: CUBIC provides the most balanced performance for mixed traffic environments in WiFi networks, offering stable congestion windows without excessive buffering while maintaining minimal impact on UDP traffic jitter. **BBR** shows promising characteristics but may require tuning specifically for WiFi environments due to observed higher jitter and retransmission rates. **NewReno** demonstrates clear bufferbloat symptoms and is not recommended for modern deployments with latency-sensitive traffic requirements.

4.7 L2→L4 Interactions in Analysis Data

This section analyzes the complex interactions between Layer 2 (MAC/PHY) and Layer 4 (Transport) across different network conditions. The analysis reveals how wireless channel characteristics propagate through the protocol stack to impact application performance.

4.7.1 Distance = 40m - STA2 BK Traffic (Strongest Case)

This scenario demonstrates the most pronounced L2→L4 interaction observed in the simulations, where background (BK) traffic shows significant performance degradation due to wireless channel conditions.

Evidence Chain of L2→L4 Impact

1. **PHY Layer (L1/L2):**
   - PHY Loss Rate: 3.07%
   - RX Drops: 30,360 packets
2. **MAC Layer (L2):**
   - Queue buildup due to retransmission attempts

TCP Variant	Suitability	Key Characteristics for Mixed Traffic
CUBIC	**Highest**	• Stable CWND without excessive buffering • Minimal impact on UDP jitter • Zero retransmissions • Most balanced performance overall
BBR	**Moderate**	• Higher UDP jitter observed • More retransmissions for lower-priority traffic • May need tuning for WiFi environments • Could be optimized with BBRv2 or rate limiting
NewReno	**Low**	• Shows bufferbloat symptoms • Unrealistically large congestion windows • May perform poorly with limited buffers • Not recommended for modern WiFi deployments

Table 4.15: TCP Variant Suitability for Mixed Traffic Environments

- Increased medium access contention

3. **TCP Layer (L4):**
   - RTT increases from 5.10ms to 22.13ms (4.3× increase!)
   - Throughput decreases from 11.52 Mbps to 9.69 Mbps (16% drop)
   - Retransmissions increase from 4 to 30 (7.5× increase!)

Cross-Layer Correlation Proof

Interpretation: At 40m distance, wireless channel errors cause MAC layer retransmissions, leading to queue buildup which directly inflates TCP RTT and triggers TCP con-

Correlation	Value	Interpretation
Queue-RTT	0.877	Strong positive correlation
Loss-CWND	0.576	Moderate positive correlation

Table 4.16: Cross-Layer Correlations at 40m Distance

gestion control mechanisms. The strong queue-RTT correlation (0.877) provides definitive evidence of L2→L4 interaction.

4.7.2 Asymmetric Distance (10m/20m) - Early Signs

This analysis examines early signs of L2→L4 interactions under moderate distance differentials, where STA2 is positioned farther (20m) than STA1 (10m).

Performance Impact Summary

Metric	STA2 (20m)	Change
PHY Loss Rate	0.81%	Near doubling (from 0.46%)
TCP BK RTT	5.27 ms	Increase from 5.10 ms
TCP BK Retransmissions	5	Increase from 4
UDP VI Jitter	0.118 ms	24% increase (from 0.095 ms)

Table 4.17: Early L2→L4 Interaction Signs at 20m Distance

Key Insight: Even moderate distance increases begin affecting transport layer performance, with UDP jitter showing particular sensitivity (24% increase) due to its real-time nature.

4.7.3 TCP Variant Analysis - BBR Sensitivity

The analysis reveals distinct L2→L4 interaction patterns across different TCP variants, with BBR showing unique sensitivity characteristics.

Metric	BBR	CUBIC	NewReno
BE Retransmissions	31	0	0
VO Jitter	0.025 ms	0.012 ms	0.011 ms
Queue-RTT Correlation	Weak	Moderate	Strong

Table 4.18: TCP Variant Sensitivity to L2→L4 Interactions

BBR-WiFi Interaction Mechanism

1. **BBR sends probe bursts** to estimate available bandwidth
2. **WiFi MAC schedules** these bursts alongside high-priority UDP traffic
3. **Collisions occur** between probe bursts and scheduled transmissions
4. **L2 retransmissions** are triggered for collided packets
5. **TCP interprets losses** as congestion, entering recovery phase
6. **Increased retransmissions** result from this interaction pattern

4.7.4 The L2→L4 Impact Cascade

This section outlines the systematic progression of channel degradation through the protocol stack, demonstrating how physical layer impairments cascade to application performance degradation.

Channel Degradation Path

1. **Distance Increase** → Signal-to-Noise Ratio (SNR) Decrease
2. **Lower SNR** → Higher Packet Error Rate (PER)
3. **Higher PER** → Increased MAC Layer Retransmissions
4. **MAC Retries** → Extended Queueing Delays at L2
5. **Queue Buildup** → Increased Round-Trip Time (RTT)
   - TCP interprets increased RTT as network congestion
6. **TCP Congestion Response:**
   - Reduces Congestion Window (CWND) → Throughput drops
   - Enters fast recovery mode → Retransmissions increase
   - May experience timeouts → Severe throughput collapse

Analysis Data Progression

Distance	PHY Loss	TCP BK RTT	TCP BK Throughput	Queue-RTT Corr.
10m	0.46%	5.10 ms	11.52 Mbps	0.005 (weak)
20m	0.81%	5.27 ms	11.52 Mbps	0.129 (weak)
40m	3.07%	22.13 ms	9.69 Mbps	0.877 (strong)

Table 4.19: L2→L4 Impact Cascade Across Distances

Critical Finding: The correlation strength (0.877) at 40m provides empirical proof that L2 queue dynamics directly determine TCP RTT performance, confirming the L2→L4 impact cascade hypothesis.

4.7.5 Where L2 Impact Becomes Dominant

Analysis of simulation data reveals a critical threshold where L2 impairments begin to dominate transport layer performance.

Performance Threshold Analysis

Performance Characteristic	**Below 1% PHY Loss**	**Above 3% PHY Loss**
TCP Throughput	Maintained at 11.52 Mbps	BK: 16% reduction (9.69 Mbps)
TCP RTT	Stable (5.10-5.32 ms)	BK: 4.3× increase (22.13 ms)
Retransmissions	Minimal (0-4)	BK: 7.5× increase (30)
L4 Awareness of L2 Issues	Minimal	Queue-RTT correlation: 0.877
Transport Layer Impact	L4 mostly unaware of L2 issues	L2 queues dominate TCP performance

Table 4.20: L2 Dominance Threshold Based on PHY Loss Rate

Threshold Implications for Network Design

- **Below 1% PHY Loss:**
  - Transport layer operates with minimal awareness of L2 impairments
  - QoS mechanisms effectively isolate traffic classes https://www.overleaf.comoverleaf
  - TCP variants show similar performance characteristics
- **Above 3% PHY Loss:**
  - L2 queue dynamics become the dominant performance factor
  - Traditional TCP congestion control may misinterpret wireless loss
  - QoS differentiation becomes challenging due to cross-layer coupling
  - Advanced techniques (e.g., cross-layer optimization) may be required

Key Insight: The 3% PHY loss rate emerges as a critical threshold where wireless channel conditions transition from being a secondary consideration to the primary determinant of transport layer performance in WiFi networks.

Chapter 5

Limitations of the Simulation Study

This chapter outlines the limitations of the current simulation study, acknowledging the simplifications and constraints that were necessary for focused analysis. These limitations provide context for interpreting the results and suggest directions for future work.

5.1 PHY Layer Simplifications

- **Simplified PHY Model:** The simulation does not include advanced PHY features such as Multiple Input Multiple Output (MIMO), beamforming, or spatial multiplexing, which are characteristic of modern 802.11ac implementations.
- **Fixed Data Rate:** The study uses a fixed data rate configuration without dynamic rate adaptation algorithms (e.g., MinstrelHT), limiting the analysis of rate adaptation's impact on QoS performance.
- **Perfect Channel Estimation:** The simulation assumes perfect channel state information without modeling channel estimation errors, which are common in real wireless environments.
- **Single Channel Operation:** The analysis is limited to single-channel operation without considering channel bonding or multi-channel operations available in 802.11ac.

5.2 MAC Layer Limitations

- **No Retry Limits:** The simulation uses default ns-3 retry mechanisms without modeling the full complexity of retry limits and backoff algorithms present in commercial implementations.
- **Simplified Queue Management:** While FQ-CoDel is implemented at the AP, the per-station queue management is simplified and may not capture all real-world queue behaviors.
- **Fixed EDCA Parameters:** The 802.11e EDCA parameters are set to default values and not optimized for specific traffic patterns or network conditions.

5.3 Network Topology Constraints

- **Two Stations Only:** The study is limited to two wireless stations, which restricts scalability analysis and may not represent behavior in dense deployment scenarios.
- **Single AP Configuration:** All simulations use a single access point without considering multi-AP deployments, handovers, or interference between adjacent cells.
- **No External Interference:** The simulation environment is interference-free, omitting the impact of co-channel interference, adjacent channel interference, or non-WiFi interference sources.
- **Simplified Mobility Models:** Mobility is modeled using basic random walk patterns without considering complex real-world mobility patterns or obstacles.

5.4 Traffic Pattern Limitations

Limitation	Description	Impact on Results
Independent Traffic Flows	No interaction between traffic generators	May underestimate contention effects
Constant Bit Rate UDP	Simplified traffic patterns	Doesn't capture bursty traffic behavior
Infinite Source Data	TCP sources never starve	May not reflect real application behavior
Fixed Traffic Mix	Static ratio of traffic classes	Limited generalization to dynamic mixes

Table 5.1: Traffic Pattern Limitations and Their Impacts

5.5 Measurement and Analysis Constraints

- **Statistical Significance:** While sample sizes are generally sufficient, certain rare events or edge cases may not be fully captured.
- **Cross-Layer Correlation Limitations:** The correlation analysis assumes linear relationships and may miss non-linear interactions.
- **Simulation Duration:** 300-second simulations may not capture long-term behaviors or network convergence patterns.

5.6 ns-3 Implementation Specifics

- **Default ns-3 Models:** The study relies on default ns-3 models and parameters, which may differ from real hardware implementations.

- **Timing Precision:** The discrete event simulation approach has inherent timing limitations that may affect microsecond-level measurements.
- **Memory and Processing Constraints:** Simulation scalability is limited by computational resources, restricting the size and complexity of scenarios.
- **Platform Version:** The study uses NS-3.44, which may have different characteristics from newer versions of the simulator.

5.6.1 Generalizability Considerations

Simulation Condition	Generalizability to Real Networks
Simplified PHY without MIMO/beamforming	Results may be conservative for modern APs
No external interference	Throughput may be overestimated
Two-station topology	May not scale to enterprise deployments
Perfect channel estimation	Latency may be underestimated
Fixed data rates	May not capture rate adaptation benefits

Table 5.2: Generalizability of Simulation Results

5.7 For Future Work

1. **Enhanced PHY Models:** Incorporate MIMO, beamforming, and dynamic rate adaptation algorithms to better represent modern 802.11ac/ax implementations.
2. **Scalability Studies:** Extend the analysis to larger numbers of stations (10+, 50+) to understand performance in dense deployment scenarios.
3. **Real-World Traffic Patterns:** Use more realistic traffic models including bursty traffic, variable bit rates, and application-specific patterns.
4. **Multi-AP Deployments:** Study interference, handovers, and load balancing in multi-AP environments.
5. **Advanced QoS Mechanisms:** Investigate newer QoS enhancements including 802.11e enhancements and application-aware scheduling.
6. **Hardware Validation:** Validate simulation results against real hardware testbeds to calibrate and verify model accuracy.

Summary: While these limitations constrain the direct applicability of results to all real-world scenarios, they enable focused analysis of specific cross-layer interactions. The study provides valuable insights into fundamental mechanisms while establishing a framework for more comprehensive investigations.

Future work will involve extending the analysis to include additional TCP variants, interference scenarios, and heterogeneous traffic mixes to further validate the robustness of the QoS framework.